

Python & SQLite Databases

Learn how to store and manage data with
Python!



Why Do We Need Databases?

Databases help us keep lots of information safe and organized — like a digital filing cabinet!

- Store thousands of records neatly
- Find information quickly
- Keep data safe even when the app closes





Meet SQLite!

SQLite is a tiny database that lives right inside your Python programs!

- No setup needed — just import and go!
- Stores everything in one small file
- Perfect for learning and small projects

How Does Python Talk to SQLite?

Python uses special commands to work with SQLite databases.

- We can Add new data
- We can Read existing data
- We can Update or Delete data too!





Setting Up SQLite in Python

Just 3 simple steps to get started!

- Step 1: import sqlite3
- Step 2: Connect to a database file
- Step 3: Create a cursor to run commands

Creating a Database & Table

A table is like a spreadsheet inside your database.

- CREATE TABLE makes a new table
- We add columns like name, age, class
- Each row is one record (like one student)





Adding Data (Insert)

Use INSERT to add new records to your table!

```
cursor.execute("""  
    INSERT INTO students  
    (name, age, class)  
    VALUES ("Ananya", 10, "5A")  
    """)
```

This adds a new student named Ananya to our database!

Reading Data (Select)

Use SELECT to read and display data from your table!

- SELECT * FROM students shows all records
- You can also pick specific columns
- Results come back as a list

```
cursor.execute("SELECT * FROM students")
results = cursor.fetchall()
for row in results:
    print(row)
```




Updating Data

Use UPDATE to change existing records in your table!

- UPDATE changes data that already exists
- Use WHERE to pick which record to change
- Always save changes with commit()

```
cursor.execute("""
    UPDATE students
    SET age = 11
    WHERE name = "Ananya"
""")
```

Deleting Data

Use DELETE to remove records you no longer need.

- DELETE FROM removes a record
- Use WHERE to pick which one to delete
- Be careful — deleted data is gone!

```
cursor.execute("""  
    DELETE FROM students  
    WHERE name = "Ananya"  
""")
```





Quick Quiz!

Test what you have learned so far!

- Q1: What does INSERT do?
- Q2: Which command reads data?
- Q3: True or False — SQLite is big and hard to use?

Why Learn Python & Databases?

Databases are everywhere — learning them now gives you superpowers!

- Build games and apps that save info
- Solve problems with data
- Get ready for cool tech jobs!





Where is Python + SQLite Used?

Python and SQLite are used in many real things you already know!

- Mobile apps that save your progress
- School systems that store grades
- Games that remember your score
- Websites that keep your login info

What Next? Practice Ideas

Try these fun projects to practice what you learned!

- Make a database of your favourite books
- Store your friends' birthdays
- Create a quiz score tracker
- Build a mini library app!



Thank You!

Thanks for learning Python & SQLite with us!
Any questions?

Keep Coding!

